



Algorithm Design II

Algorithms with Numbers I

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Two Seemingly Similar Problems



Factoring: Given a number N , express it as a product of its prime factors.

Primality: Given a number N , determine whether it is a prime.

We believe that **Factoring** is hard and much of the electronic commerce is built on this assumption.

There are efficient algorithms for **Primality**, e.g., **AKS** test by Manindra Agrawal, Neeraj Kayal, and Nitin Saxena.

A Notable Result



The **AKS primality test** is a deterministic primality-proving algorithm created and published by **Manindra Agrawal**, **Neeraj Kayal**, and **Nitin Saxena**, computer scientists at the Indian Institute of Technology Kanpur, on August 6, 2002. The algorithm was the first to determine whether any given number is prime or composite within polynomial time. The authors received the 2006 **Gödel Prize** and the 2006 **Fulkerson Prize** for this work.

Preliminaries

How to Represent Numbers



We are most familiar with **decimal** representation:

- 1024

But computers use **binary** representation:

- $1 \underbrace{0 \dots 0}_{10 \text{ times}}$

The bigger the **base** is, the shorter the **representation** is. But how much do we really gain by choosing large base?



Q: How many digits are needed to represent the number $N \geq 0$ in base b ?

$$\lceil \log_b(N + 1) \rceil$$

Q: How much does the size of a number change when we change bases?

$$\log_b N = \frac{\log_a N}{\log_a b}$$

In O notation, the base is irrelevant, and thus we write the size simply as $O(\log N)$

Roles of $\log N$



$\log N$ is the power to which you need to **raise 2** in order to obtain N .

It can also be seen as the number of times you must **halve N** to get down to **1**. (More precisely: $\lceil \log N \rceil$.)

It is the number of bits in the binary representation of N . (More precisely: $\lceil \log (N + 1) \rceil$.)

It is also the depth of a **complete binary tree** with N nodes. (More precisely: $\lceil \log N \rceil$.)

It is even the sum $1 + 1/2 + 1/3 + \dots + 1/n$, to within a constant factor.

Basics Arithmetic



Lemma

The sum of any three single-digit number is at most two digits long.

Addition



This rule holds not just in decimal but in any $b \geq 2$.

In binary, the maximum possible sum of three single-bit numbers is 3, which is a 2-bit number.

This simple rule gives us a way to add two numbers in any bases.

$$\begin{array}{rcccccc} & & 1 & & & 1 & 1 & 1 \\ & & & & 1 & 1 & 0 & 1 & 0 & 1 \\ & & & & 1 & 0 & 0 & 0 & 1 & 1 \\ \hline 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \end{array}$$



Q: Given two binary number x and y , how long does our algorithm take to add them?

The answer expressed as a **function of the size of the input**: the number of bits of x and y (suppose they are n bit long).

The sum of x and y is $n + 1$ bits at most. Each individual bit of this sum gets computed in a **fixed amount** of time.

The total running time for the addition is of form $c_0 + c_1 n$, where c_0 and c_1 are some constants, i.e., $O(n)$.



Q: Can we do better?

In order to add two n -bit numbers, we must read them and write down the answer, and even that requires n operations.

So the addition algorithm is **optimal**.

Perform Addition in One Step?



A single instruction we can add integers whose size in bits is within the word length of today's computer - 64 perhaps.

It is often useful and necessary to handle numbers much larger than this, perhaps several thousand bits long.

To study the basic algorithms encoded in the hardware of today's computers, we shall focus on the bit complexity of the algorithm, the number of elementary operations on individual bits.

Multiplication



The grade-school algorithm for multiplying two number x and y is to create an array of intermediate sums.

If x and y are both n bit, then there are n intermediate rows with length of up to $2n$ bit.
(Q: why?)

$$\begin{array}{r} \\ \\ \\ \\ \\ \\ + \\ \hline 1 \end{array}$$

$$\underbrace{O(n) + \dots + O(n)}_{n-1}$$
$$O(n^2)$$

Quiz



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What is the complexity of a number times 2?

Multiplication by Al Khwarizmi



- write them next to each other.
- **halve** the first number by **2**, dropping the **.5**, and **double** the second number.
- keep going till the first number **gets down** to 1.
- **strike out** all the rows where the first number is **even**.
- **add up** the remains in the second columns.

$$\begin{array}{r} 11 \quad 13 \\ 5 \quad 26 \\ 2 \quad 52 \\ \hline 1 \quad 104 \\ \hline 143 \end{array}$$

- The left is to calculate the binary number.
- The right is to shift the row!

Multiplication in Recursion



```
MULTIPLY ( $x, y$ )  
Two  $n$ -bit integers  $x$  and  $y$ , where  $y \geq 0$ ;  
if  $y = 0$  then return 0;  
 $z$  = MULTIPLY ( $x, \lfloor y/2 \rfloor$ );  
if  $y$  is even then  
    | return  $2z$ ;  
    | else return  $x + 2z$ ;  
end
```

Another formulation:

$$x \cdot y = \begin{cases} 2(x \cdot \lfloor y/2 \rfloor) & \text{if } y \text{ is even} \\ x + 2(x \cdot \lfloor y/2 \rfloor) & \text{if } y \text{ is odd} \end{cases}$$



Multiplication in Recursion

Q: How long does the algorithm take?

It will terminate after n recursive calls, since at each call y is halved.

At each call requires these operations:

- a division by 2 (right shift);
- a test for odd/even (looking up the last bit);
- a multiplication by 2 (left shift);
- and a possibly one addition.

A total operations are $O(n)$, The total time taken is thus $O(n^2)$.

Q: Can we do better?

- Yes!



DIVIDE (x, y)

Two n -bit integers x and y , where $y \geq 1$;

if $x = 0$ then return ($0, 0$);

$(q, r) = \text{DIVIDE}(\lfloor x/2 \rfloor, y)$;

$q = 2 \cdot q, r = 2 \cdot r$;

if x is odd then $r = r + 1$;

if $r \geq y$ then $r = r - y, q = q + 1$;

return (q, r);

Q: How long does the algorithm take?

- Exercise 1.8!

Modular Arithmetic

What Is Modular



Modular arithmetic is a system for dealing with restricted ranges of integers.

x modulo N is the remainder when x is divided by N ; that is, if $x = qN + r$ with $0 \leq r < N$, then x modulo N is equal to r .

x and y are congruent modulo N if they differ by a multiple of N , i.e.

$$x \equiv y \pmod{N} \Leftrightarrow N \text{ divides } (x - y)$$

Two Interpretations



- ① It limits numbers to a predefined range $\{0, 1, \dots, N\}$ and wraps around whenever you try to leave this range - like the hand of a clock.
- ② Modular arithmetic deals with all the integers, but divides them into N equivalence classes, each of the form $\{i + k \cdot N \mid k \in \mathbb{Z}\}$ for some i between 0 and $N - 1$.

Two's Complement



Modular arithmetic is nicely illustrated in **two's complement**, the most common format for storing **signed integers**.

It uses n bits to represent numbers in the range

$$-2^{n-1}, 2^{n-1} - 1$$

and is usually described as follows:

- **Positive integers**, in the range 0 to $2^{n-1} - 1$, are stored in regular binary and have a **leading bit** of 0 .
- **Negative integers** $-x$, with $1 \leq x \leq 2^{n-1}$, are stored by first constructing x in binary, then flipping all the bits, and finally adding 1 . The **leading bit** in this case is 1 .

Two's Complement



0	1	1	1	1	1	1	1	= 127
0	0	0	0	0	0	1	0	= 2
0	0	0	0	0	0	0	1	= 1
0	0	0	0	0	0	0	0	= 0
1	1	1	1	1	1	1	1	= -1
1	1	1	1	1	1	1	0	= -2
1	0	0	0	0	0	0	1	= -127
1	0	0	0	0	0	0	0	= -128

(from wiki)

Rules



Substitution rules: if $x \equiv x' \pmod{N}$ and $y \equiv y' \pmod{N}$, then

$$\begin{aligned}x + y &\equiv x' + y' \pmod{N} \\ xy &\equiv x'y' \pmod{N}\end{aligned}$$

$x + (y + z) \equiv (x + y) + z$	$\pmod{N}$	Associativity
$xy \equiv yx$	$\pmod{N}$	Commutativity
$x(y + z) \equiv xy + xz$	$\pmod{N}$	Distributivity

It is legal to reduce intermediate results to their remainders modulo N at any stage.

$$2^{345} \equiv (2^5)^{69} \equiv 32^{69} \equiv 1^{69} \equiv 1 \pmod{31}$$



Since x and y are each in the range 0 to $N - 1$, their **sum** is between 0 and $2(N - 1)$.

If the sum exceeds $N - 1$, we merely need to **subtract off** N to bring it back into the required range.

The overall computation therefore consists of an **addition**, and possibly a **subtraction**, of numbers that never exceed $2N$.

Its running time is $O(n)$, where $n = \lceil \log N \rceil$.

Modular Multiplication



The product of x and y can be as large as $(N - 1)^2$, but this is still at most $2n$ bits long since

$$\log (N - 1)^2 = 2\log (N - 1) \leq 2n$$

To reduce the answer $\bmod N$, we compute the remainder upon dividing it by N . ($O(n^2)$)

Multiplication thus remains a **quadratic** operation.



Not quite so easy!

In ordinary arithmetic there is just one tricky case - **division by zero**.

It turns out that in modular arithmetic there are potentially other such cases as well.

Whenever division is legal, however, it can be managed in **cubic time**, $O(n^3)$.

Modular Exponentiation



In the **cryptosystem**, it is necessary to compute $x^y \pmod N$ for values of x , y , and N that are **several hundred bits** long.

The result is some number $\pmod N$ and is therefore a few hundred bits long. However, the raw value x^y could be much, much longer.

When x and y are just 20-bit numbers, x^y is at least

$$(2^{19})^{(2^{19})} = 2^{(19)(524288)}$$

about **10 million bits** long!

Modular Exponentiation



To make sure the numbers never grow too large, we need to perform all intermediate computations modulo N .

First idea: calculate $x^y \bmod N$ by repeatedly multiplying by x modulo N .

The resulting sequence of intermediate products,

$$x \bmod N \rightarrow x^2 \bmod N \rightarrow x^3 \bmod N \rightarrow \dots \rightarrow x^y \bmod N$$

consists of numbers that are smaller than N , and so the individual multiplications do not take too long.

But imagine if y is 500 bits long . . .

Modular Exponentiation



Second idea: starting with x and squaring repeatedly modulo N , we get

$$x \bmod N \rightarrow x^2 \bmod N \rightarrow x^4 \bmod N \rightarrow x^8 \bmod N \rightarrow \dots x^{2^{\lfloor \log y \rfloor}} \bmod N$$

Each takes just $O(\log^2 N)$ time to compute, and in this case there are only $\log y$ multiplications.

To determine $x^y \bmod N$, multiply together an **appropriate subset** of these powers, those corresponding to **1**'s in the binary representation of y .

For instance,

$$x^{25} = x^{11001_2} = x^{10000_2} \cdot x^{1000_2} \cdot x^{1_2} = x^{16} \cdot x^8 \cdot x^1$$

Modular Exponentiation



MODEXP (x, y, N)

Two n -bit integers x and N , and an integer exponent y ;

if $y = 0$ **then** return 1;

z = MODEXP ($x, \lfloor y/2 \rfloor, N$);

if y *is even* **then**

 return $z^2 \bmod N$;

else return $x \cdot z^2 \bmod N$;

end

Another formulation:

$$x^y \bmod N = \begin{cases} (x^{\lfloor y/2 \rfloor})^2 \bmod N & \text{if } y \text{ is even} \\ x \cdot (x^{\lfloor y/2 \rfloor})^2 \bmod N & \text{if } y \text{ is odd} \end{cases}$$

Modular Exponentiation



MODEXP (x, y, N)

Two n -bit integers x and N , and an integer exponent y ;

if $y = 0$ **then** return 1;

$z = \text{MODEXP}(x, \lfloor y/2 \rfloor, N)$;

if y *is even* **then**

 return $z^2 \bmod N$;

else return $x \cdot z^2 \bmod N$;

end

The algorithm will halt after at most n recursive calls, and during each call it multiplies n -bit numbers. for a total running time of $O(n^3)$

Euclid's Algorithm for Greatest Common Divisor



Q: Given two integers x and y , how to find their **greatest common divisor** ($\gcd(x, y)$)?

Euclid's rule

If x and y are positive integers with $x \geq y$, then $\gcd(x, y) = \gcd(x \bmod y, y)$.

Proof:

It is enough to show the rule $\gcd(x, y) = \gcd(x - y, y)$. Result can be derived by repeatedly subtracting y from x .

Euclid's Algorithm for Greatest Common Divisor



EUCLID(x, y)

Two integers x and y with $x \geq y$;

if $y = 0$ **then** return x ;

return (EUCLID($y, x \bmod y$));

Lemma

If $a \geq b \geq 0$, then $a \bmod b < a/2$

Proof:

- if $b \leq a/2$, $a \bmod b < b \leq a/2$;
- if $b > a/2$, $a \bmod b = a - b < a/2$.

Euclid's Algorithm for Greatest Common Divisor



EUCLID(x, y)

Two integers x and y with $x \geq y$;

if $y = 0$ **then** return x ;

return (EUCLID($y, x \bmod y$));

Lemma

If $a \geq b \geq 0$, then $a \bmod b < a/2$

This means that after any two **consecutive rounds**, both arguments, x and y are at the very least **halved** in value, i.e., the length of each decreases at least one bit.

Euclid's Algorithm for Greatest Common Divisor



EUCLID(x, y)

Two integers x and y with $x \geq y$;

if $y = 0$ **then** return x ;

return (EUCLID($y, x \bmod y$));

Lemma

If $a \geq b \geq 0$, then $a \bmod b < a/2$

If they are initially n -bit integers, then the base case will be reached within $2n$ recursive calls. Since each call involves a quadratic-time division, the total time is $O(n^3)$.

An Extension of Euclid's Algorithm



Q: Suppose someone claims that d is the **greatest common divisor** of x and y , how can we check this?

It is not enough to verify that d divides both x and y ...

Lemma

If d divides both x and y , and $d = ax + by$ for some integers a and b , then necessarily $d = \gcd(x, y)$.

Proof:

$d \leq \gcd(x, y)$, obviously;

$d \geq \gcd(x, y)$, since $\gcd(x, y)$ can divide x and y , it must also divide $ax + by = d$.

An Extension of Euclid's Algorithm



EXTENDED-EUCLID (x, y)

Two integers x and y with $x \geq y \geq 0$;

if $y = 0$ **then** return ($1, 0, x$);

$(x', y', d) = \text{EXTENDED-EUCLID}(y, x \bmod y)$;

return ($y', x' - \lfloor x/y \rfloor y', d$);

Correctness of the algorithm?

DIY!

Modular Inverse



We say x is the **multiplicative inverse** of $a \bmod N$ if

$$ax \equiv 1 \bmod N$$

There can be at most one such $x \bmod N$, denoted a^{-1} .

Remark: The inverse does not always exist! for instance, 2 is not invertible modulo 6.



Lemma

If $\gcd(a, N) > 1$, then $ax \not\equiv 1 \pmod N$.

Proof:

$ax \pmod N = ax + kN$, then $\gcd(a, N)$ divides $ax \pmod N$

If $\gcd(a, N) = 1$, then extended Euclid algorithm gives us integers x and y such that $ax + Ny = 1$, which means $ax \equiv 1 \pmod N$. Thus x is a 's sought inverse.



Theorem (Modular Division Theorem)

For any $a \bmod N$, a has a multiplicative inverse modulo N if and only if it is relatively prime to N .

When the inverse exists, it can be found in time $O(n^3)$ by running the extended Euclid algorithm.

This resolves the issues of **modular division**: when working modulo N , can divide by numbers **relatively prime** to N . And to actually carry out the division, multiply by the **inverse**.